STATIC TIMING ANALYSIS

Multi-Corner Timing Results, Graphs and Evaluation

Flow: OpenSTA | Platform: SkyWater SKY130 HD | Analysis: PVT corner comparison

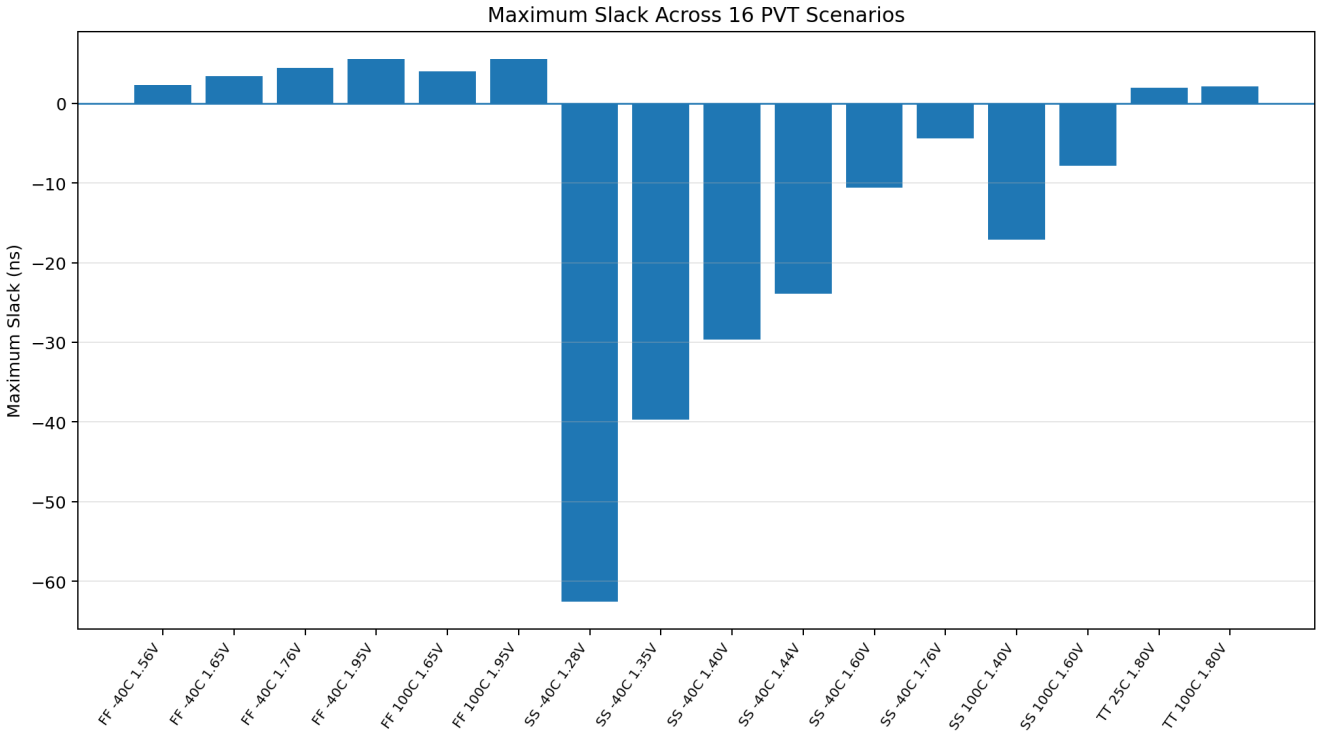
Metric	Observed result
PVT scenarios shown	16
Scenarios marked MET	8
Scenarios marked VIOLATED	8
Worst maximum slack	–62.56 ns (SS, –40°C, 1.28 V)

Executive summary. The supplied table contains 16 PVT scenarios. All displayed FF and TT rows are marked MET, while all displayed SS rows are marked VIOLATED due to negative maximum slack. The worst reported maximum slack is –62.56 ns at SS, –40°C, 1.28 V. Graphs in this report make the corner-level differences and SS voltage trend easier to compare.

Scope: analysis is based on the supplied screenshot/table and terminal screenshots. Review complete timing reports, constraints, and path details before sign-off.

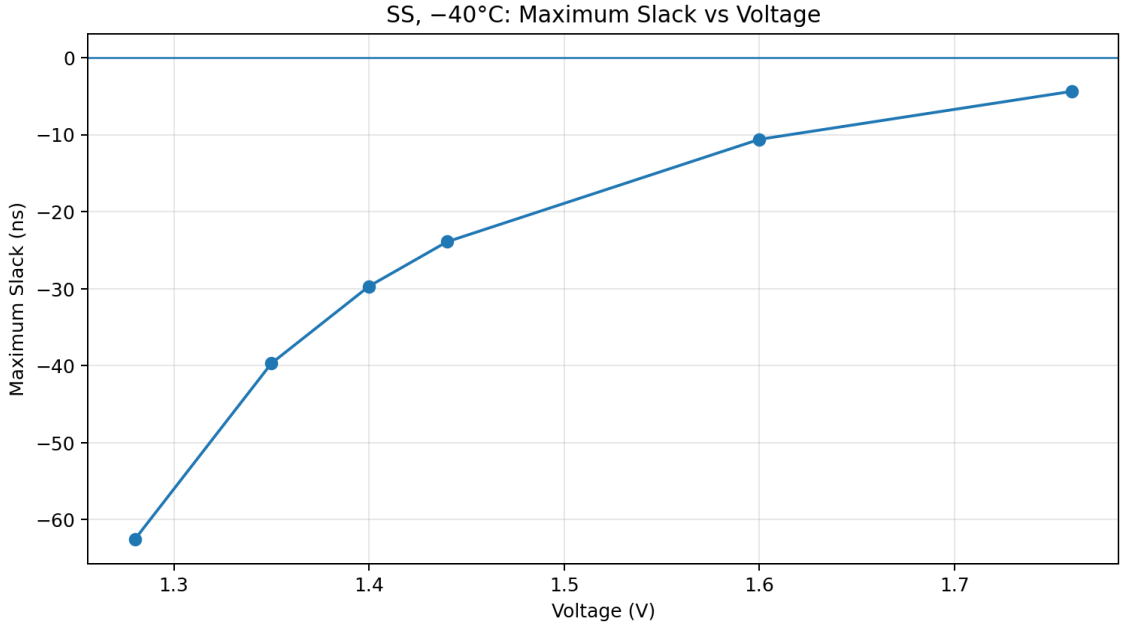
1. Maximum Slack by PVT Scenario

Values below zero indicate negative maximum slack. The graph shows a pronounced negative-slack cluster in the SS scenarios; FF and TT scenarios remain positive.



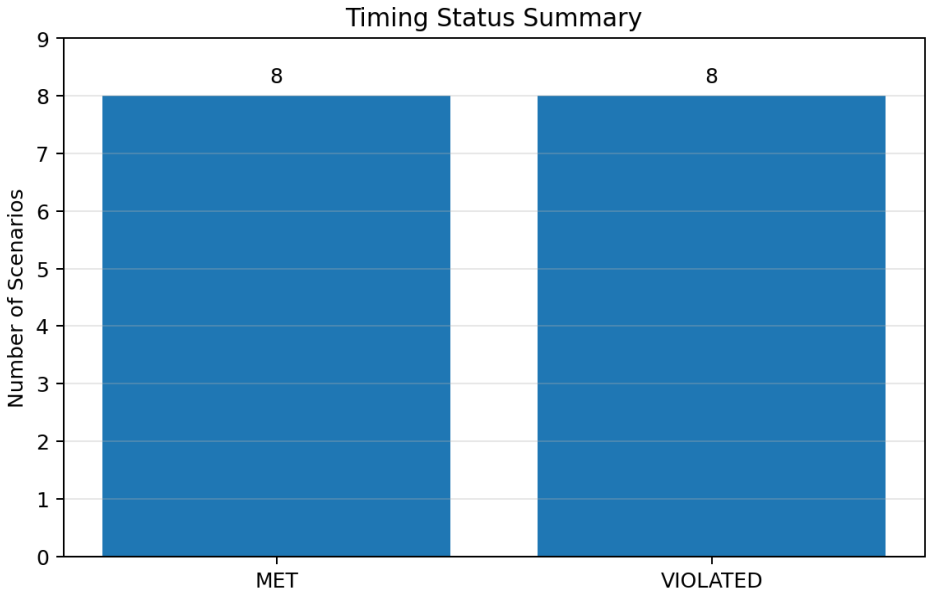
2. SS Corner Voltage Sweep at −40°C

Across the six shown SS / −40°C cases, maximum slack becomes less negative as voltage rises from 1.28 V to 1.76 V. All six points remain below zero, so this trend does not indicate timing closure.



3. Timing Status Summary

The table has 8 MET and 8 VIOLATED scenarios. This count is based on the status column in the supplied image.



4. Results Table

#	Process	Temp	Voltage	Min Slack (ns)	Max Slack (ns)	Status
1	FF	–40°C	1.56 V	0.29	2.28	MET
2	FF	–40°C	1.65 V	0.26	3.41	MET
3	FF	–40°C	1.76 V	0.22	4.43	MET
4	FF	–40°C	1.95 V	0.19	5.59	MET
5	FF	100°C	1.65 V	0.25	4.05	MET
6	FF	100°C	1.95 V	0.20	5.57	MET
7	SS	–40°C	1.28 V	1.83	–62.56	VIOLATED
8	SS	–40°C	1.35 V	1.35	–39.70	VIOLATED
9	SS	–40°C	1.40 V	1.12	–29.69	VIOLATED
10	SS	–40°C	1.44 V	0.99	–23.91	VIOLATED
11	SS	–40°C	1.60 V	0.66	–10.61	VIOLATED
12	SS	–40°C	1.76 V	0.50	–4.38	VIOLATED
13	SS	100°C	1.40 V	0.91	–17.12	VIOLATED
14	SS	100°C	1.60 V	0.64	–7.87	VIOLATED
15	TT	25°C	1.80 V	0.31	1.94	MET
16	TT	100°C	1.80 V	0.31	2.09	MET

5. Findings and Interpretation

FF corner: Six scenarios are MET; maximum slack ranges from 2.28 ns to 5.59 ns.

SS corner: Eight scenarios are VIOLATED. At –40°C, the voltage sweep improves maximum slack from –62.56 ns (1.28 V) to –4.38 ns (1.76 V), but it remains negative.

TT corner: Both scenarios are MET, with maximum slack of 1.94 ns at 25°C and 2.09 ns at 100°C.

The slow-slow corner appears to be the primary timing concern. The negative slack suggests that one or more reported timing requirements are not met in those cases. Do not infer a fix from voltage alone; use approved operating limits and inspect the critical timing paths and constraints.

6. Recommended Next Steps

1. Inspect the worst SS path (–40°C, 1.28 V): startpoint, endpoint, cell delay, net delay, and required time.
2. Validate SDC clocks, clock uncertainty, input/output delays, generated clocks, false paths, and multicycle paths.
3. Verify that Liberty libraries and RC/parasitic files match each PVT corner.
4. Compare the same critical path across SS, TT, and FF to isolate cell-delay, interconnect, or constraint effects.
5. Re-run all required corners after optimization; include setup, hold, and unconstrained-path checks.

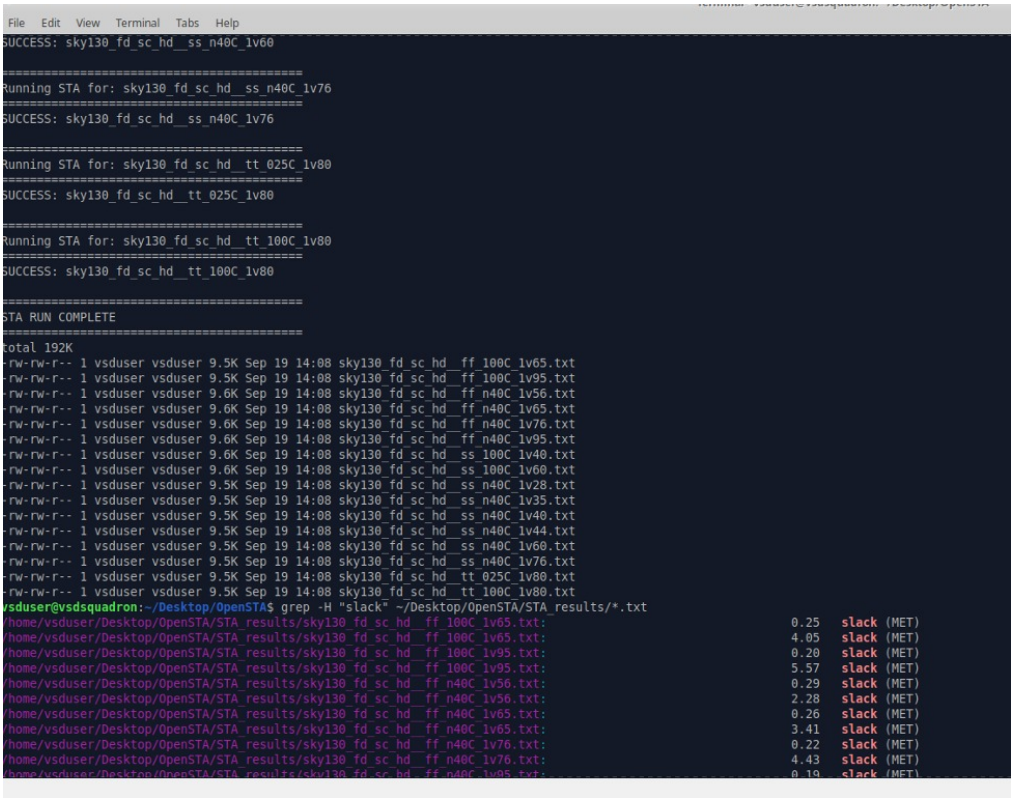
7. OpenSTA Run Evidence

The following screenshots document run completion, generated result files, slack extraction, and the source timing summary.

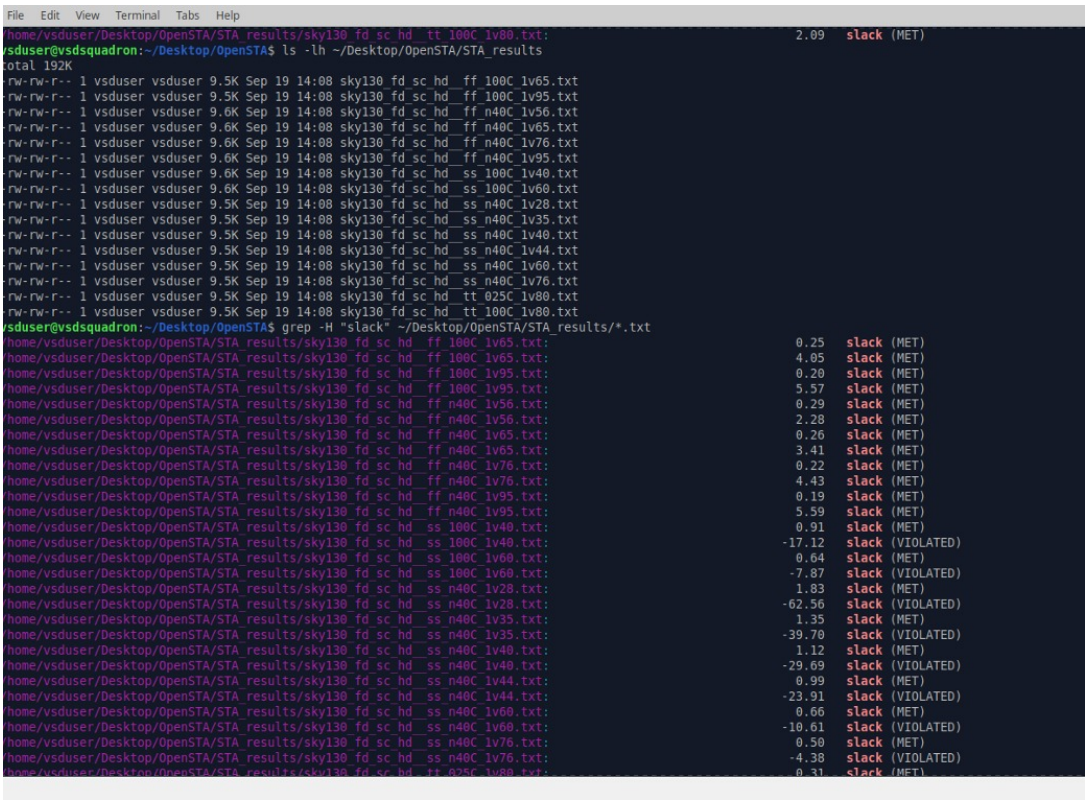
Pre_STA_analysis.jpeg

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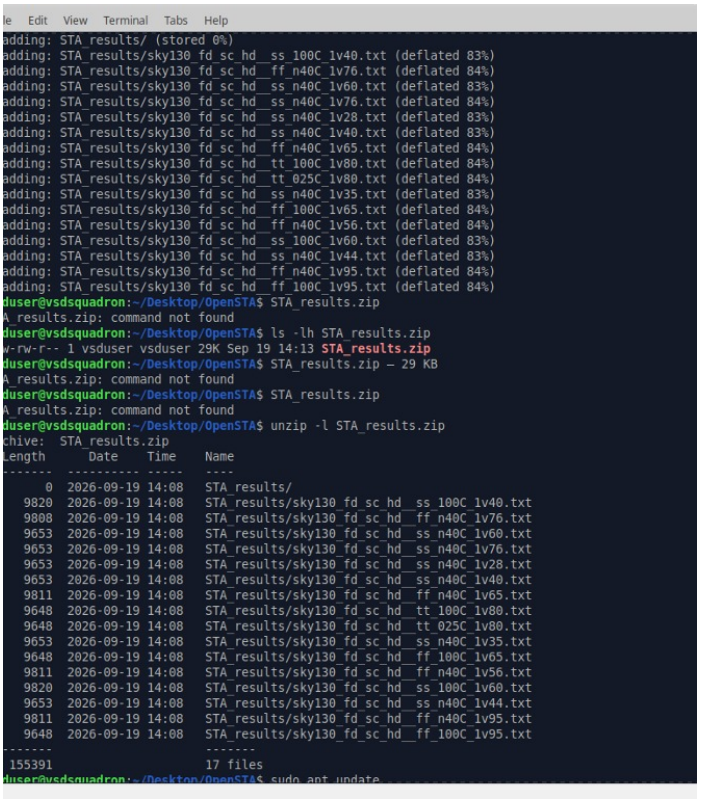
STA_1.jpeg



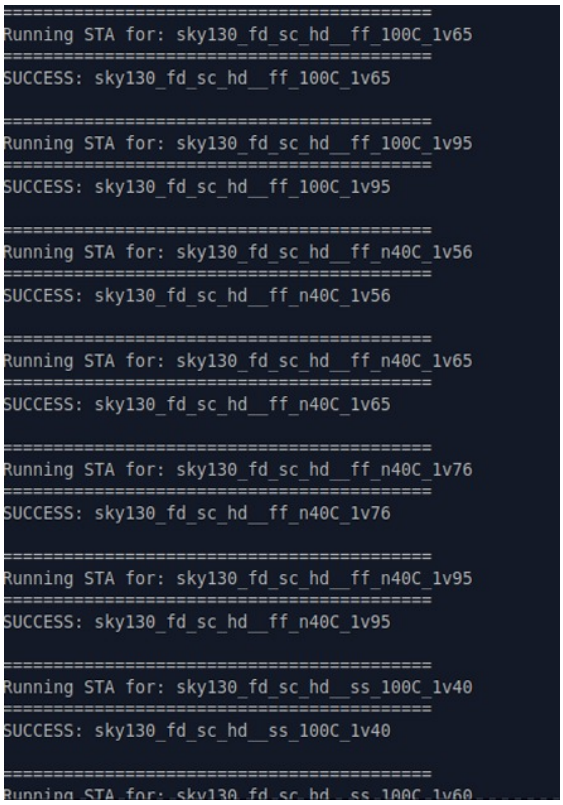
STA_2.jpeg



STA_3.jpeg



STA_diff_corners.jpeg


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Running STA for: sky130_fd_sc_hd__ff_100C_1v65
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SUCCESS: sky130_fd_sc_hd__ff_100C_1v65
==========
Running STA for: sky130_fd_sc_hd__ff_100C_1v95
==========
SUCCESS: sky130_fd_sc_hd__ff_100C_1v95
==========
Running STA for: sky130_fd_sc_hd__ff_n40C_1v56
==========
SUCCESS: sky130_fd_sc_hd__ff_n40C_1v56
==========
Running STA for: sky130_fd_sc_hd__ff_n40C_1v65
==========
SUCCESS: sky130_fd_sc_hd__ff_n40C_1v65
==========
Running STA for: sky130_fd_sc_hd__ff_n40C_1v76
==========
SUCCESS: sky130_fd_sc_hd__ff_n40C_1v76
==========
Running STA for: sky130_fd_sc_hd__ff_n40C_1v95
==========
SUCCESS: sky130_fd_sc_hd__ff_n40C_1v95
==========
Running STA for: sky130_fd_sc_hd__ss_100C_1v40
==========
SUCCESS: sky130_fd_sc_hd__ss_100C_1v40
==========
Running STA for: sky130_fd_sc_hd__ss_100C_1v60